Powers of Ten

Times 10: every digit moves one place to the left.

34 × 10 = 340

thousands	hundreds	tens	ones		tenths	hundredths	thousandths
		3	4	.			
	3	4	0	.			

Each digit is worth 10 times as much, so it moves one place left.
The empty ones place gets a 0.

4.6 ÷ 10 = 0.46

			4	. 6		
			0	. 4	6	

Divide by 10: one place right.
Divide by 100: two places right.

Powers of ten: how many 10s you multiply

$10^2 = 10 \times 10 = 100$

$10^3 = 10 \times 10 \times 10 = 1{,}000$

$10^4 = 10{,}000$

The small number says how many 10s, and how many zeros.

Your turn!

$5.2 \times 100 =$ 520

For teachers

Goal: multiply and divide by 10, 100 and 1,000 by moving digits in the place-value chart, and powers of ten (5.NBT.1–2). Say “The digits move; the point stays.” Never “add a zero”.

Powers of Ten: Practice

Times: digits move left. Divide: digits move right.

Part 1 Move the digits Write the new number in the second row.

1)

th	h	t	o	tth	hth	thth
		2	7			
	2	7	0			

$27 \times 10 =$ **270**

2)

th	h	t	o	tth	hth	thth
			6	4		
			0	6	4	

$6.4 \div 10 =$ **0.64**

3)

th	h	t	o	tth	hth	thth
			3	1	5	
	3	1	5			

$3.15 \times 100 =$ **315**

4)

th	h	t	o	tth	hth	thth
	8	2	0			
			8	2		

$820 \div 100 =$ **8.2**

5)

th	h	t	o	tth	hth	thth
			0	9		
			9			

$0.9 \times 10 =$ **9**

Powers of Ten: Practice (continued)

Part 1 Move the digits (continued)

6)

th	h	t	o		tth	hth	thth
	4	0	5	.			
		4	0	.	5		

$405 \div 10 =$ 40.5

Part 2 Times 10, 100 and 1,000 No help this time.

7) $47 \times 100 =$ 4,700

8) $3.6 \times 10 =$ 36

9) $0.25 \times 100 =$ 25

10) $8 \times 1{,}000 =$ 8,000

11) $5.04 \times 100 =$ 504

12) $62 \times 10 =$ 620

13) $0.7 \times 1{,}000 =$ 700

14) $19 \times 1{,}000 =$ 19,000

15) $2.38 \times 10 =$ 23.8

Powers of Ten: Practice (continued)

Part 2 Times 10, 100 and 1,000 (continued)

16) $0.06 \times 100 = 6$

17) $715 \times 10 = 7{,}150$

18) $4.5 \times 1{,}000 = 4{,}500$

19) $0.81 \times 10 = 8.1$

20) $33 \times 100 = 3{,}300$

21) $1.9 \times 100 = 190$

Part 3 Divide by 10, 100 and 1,000 No help this time.

22) $4{,}500 \div 100 = 45$

23) $36 \div 10 = 3.6$

24) $7 \div 100 = 0.07$

25) $8{,}000 \div 1{,}000 = 8$

26) $52 \div 100 = 0.52$

27) $9.4 \div 10 = 0.94$

Powers of Ten: Practice (continued)

Part 3 Divide by 10, 100 and 1,000 (continued)

28) $630 \div 10 =$ **63**

29) $1{,}250 \div 1{,}000 =$ **1.25**

30) $3 \div 10 =$ **0.3**

31) $74 \div 100 =$ **0.74**

32) $580 \div 100 =$ **5.8**

33) $2.6 \div 10 =$ **0.26**

34) $9{,}000 \div 100 =$ **90**

35) $45 \div 1{,}000 =$ **0.045**

36) $0.8 \div 10 =$ **0.08**

Part 4 Powers of ten No help this time.

37) $10^2 =$ **100**

38) $10^4 =$ **10,000**

Powers of Ten: Practice (continued)

Part 4 Powers of ten (continued)

39) $10^1 = \boxed{10}$

40) $10^5 = \boxed{100{,}000}$

41) $3 \times 10^2 = \boxed{300}$

42) $7 \times 10^3 = \boxed{7{,}000}$

43) $2.5 \times 10^2 = \boxed{250}$

44) $10^3 = \boxed{1{,}000}$

45) $6 \times 10^4 = \boxed{60{,}000}$

46) $4.2 \times 10^1 = \boxed{42}$

Part 5 Times what? No help this time. Count how many places the digits moved.

47) $52 \times \boxed{100} = 5{,}200$

48) $3.7 \times \boxed{10} = 37$

49) $0.4 \times \boxed{1{,}000} = 400$

Powers of Ten: Practice (continued)

Part 5 Times what? (continued)

50) $16 \times \boxed{1{,}000} = 16{,}000$

51) $2.09 \times \boxed{100} = 209$

52) $81 \times \boxed{10} = 810$

53) $0.35 \times \boxed{100} = 35$

54) $7 \times \boxed{100} = 700$

Part 6 Times 10, 100 and 1,000

55) $682 \times 1{,}000 = \boxed{682{,}000}$

56) $607 \times 100 = \boxed{60{,}700}$

57) $237 \times 10 = \boxed{2{,}370}$

58) $310 \times 100 = \boxed{31{,}000}$

59) $55.8 \times 100 = \boxed{5{,}580}$

60) $34.6 \times 100 = \boxed{3{,}460}$

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

61) $5.45 \times 10 = 54.5$

62) $2.15 \times 1{,}000 = 2{,}150$

63) $4.16 \times 1{,}000 = 4{,}160$

64) $52 \times 10 = 520$

65) $539 \times 1{,}000 = 539{,}000$

66) $4.96 \times 1{,}000 = 4{,}960$

67) $17.2 \times 10 = 172$

68) $3.1 \times 1{,}000 = 3{,}100$

69) $985 \times 100 = 98{,}500$

70) $264 \times 1{,}000 = 264{,}000$

71) $385 \times 10 = 3{,}850$

72) $8.32 \times 100 = 832$

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

73) $405 \times 10 =$ 4,050

74) $3.16 \times 100 =$ 316

75) $1.35 \times 1{,}000 =$ 1,350

76) $349 \times 1{,}000 =$ 349,000

77) $3.32 \times 10 =$ 33.2

78) $58 \times 100 =$ 5,800

Part 7 Divide by 10, 100 and 1,000

79) $6{,}898 \div 100 =$ 68.98

80) $245 \div 100 =$ 2.45

81) $3{,}505 \div 1{,}000 =$ 3.505

82) $7{,}911 \div 1{,}000 =$ 7.911

83) $5{,}342 \div 10 =$ 534.2

84) $3{,}504 \div 100 =$ 35.04

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

85) 7,984 ÷ 1,000 = 7.984

86) 9,978 ÷ 10 = 997.8

87) 6,099 ÷ 10 = 609.9

88) 7,192 ÷ 1,000 = 7.192

89) 6,087 ÷ 1,000 = 6.087

90) 5,447 ÷ 1,000 = 5.447

91) 3,345 ÷ 10 = 334.5

92) 5,160 ÷ 10 = 516

93) 1,694 ÷ 100 = 16.94

94) 9,459 ÷ 1,000 = 9.459

95) 3,635 ÷ 10 = 363.5

96) 9,926 ÷ 1,000 = 9.926

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

97) $1{,}450 \div 100 = \boxed{14.5}$

98) $9{,}342 \div 10 = \boxed{934.2}$

99) $3{,}067 \div 100 = \boxed{30.67}$

100) $9{,}635 \div 1{,}000 = \boxed{9.635}$

101) $5{,}628 \div 10 = \boxed{562.8}$

102) $1{,}958 \div 100 = \boxed{19.58}$

Part 8 Remember? Problems from earlier packets.

103) $0.62 \ \boxed{>} \ 0.6$

104) $4 \times \frac{7}{10} = \frac{\boxed{28}}{10}$

105) area: $18 \times 7 = \boxed{126}$ sq ft

106) $48 \times 25 = \boxed{1{,}200}$

107) $6 \text{ L} = \boxed{6{,}000} \text{ mL}$

108) $40 + \boxed{140} = 180$ degrees

Powers of Ten: Practice (continued)

Part 9 Word problems Think about how the digits move.

109) A box holds 24 pencils. How many pencils are in 100 boxes?

Number sentence **24 × 100 = 2,400**

Answer **2,400** pencils

Draw or show your work

110) A stack of 10 coins is 1.8 cm tall. How tall is one coin?

Number sentence **1.8 ÷ 10 = 0.18**

Answer **0.18** cm

Draw or show your work

111) A sticker costs $0.35. How much do 10 stickers cost?

Number sentence **0.35 × 10 = 3.5**

Answer **$3.50**

Draw or show your work

112) A farm has 4,500 apples to pack in 100 equal bags. How many apples go in each bag?

Number sentence **4,500 ÷ 100 = 45**

Answer **45** apples

Draw or show your work

113) Challenge: A paper is 0.1 mm thick. How thick is a stack of 1,000 papers, in cm? (10 mm = 1 cm)

Number sentence **0.1 × 1,000 = 100 mm = 10 cm**

Answer **10** cm

Draw or show your work

Check-In: Powers of Ten

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Move the digits

1)

th	h	t	o	tth	hth	thth
			5 .	2		
			0 .	0	5	2

$5.2 \div 100 =$ 0.052

Part 2 Times 10, 100 and 1,000

2) $13 \times 10 =$ 130

3) $9.1 \times 1{,}000 =$ 9,100

4) $740 \times 10 =$ 7,400

Part 3 Divide by 10, 100 and 1,000

5) $56 \div 10 =$ 5.6

6) $4{,}100 \div 100 =$ 41

7) $4{,}100 \div 10 =$ 410

Part 4 Powers of ten

8) $9 \times 10^3 =$ 9,000

9) $1.5 \times 10^3 =$ 1,500